Year 7 standard elaborations — Australian Curriculum v9.0: Science

ACiQ | v9.0

Purpose

The standards elaborations (SEs) support teachers to connect curriculum to evidence in assessment so that students are assessed on what they have had the opportunity to learn. The SEs can be used to:

- make consistent and comparable judgments, on a five-point scale, about the evidence of learning in a folio of student work across a year/band
- develop task-specific standards (or marking guides) for individual assessment tasks
- quality assure planning documents to ensure coverage of the achievement standard across a year/band.

Structure

The SEs have been developed using the Australian Curriculum achievement standard. The achievement standard for Science describes what students are expected to know and be able to do at the end of each year. Teachers use the SEs during and at the end of a teaching period to make on-balance judgments about the qualities in student work that demonstrate the depth and breadth of their learning.

In Queensland, the achievement standard represents the C standard — a sound level of knowledge and understanding of the content, and application of skills. The SEs are presented in a matrix where the discernible differences and/or degrees of quality between each performance level are highlighted. Teachers match these discernible differences and/or degrees of quality to characteristics of student work to make judgments across a five-point scale.

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Year 7 Australian Curriculum: Science achievement standard
By the end of Year 7 students explain how biological diversity is ordered and organised. They represent flows of matter and energy in ecosystems and predict the effects of environmental changes. They model cycles in the Earth-sun-moon system and explain the effects of these cycles on Earth phenomena. They represent and explain the effects of forces acting on objects. They use particle theory to explain the physical properties of substances and develop processes that separate mixtures. Students identify the factors that can influence development of and lead to changes in scientific knowledge. They explain how scientific responses are developed and can impact society. They explain the role of science communication in shaping viewpoints, policies and regulations. Students plan and conduct safe, reproducible investigations to test relationships and aspects of scientific models. They identify potential ethical issues and intercultural considerations required for field locations or use of secondary data. They use equipment to generate and record data with precision. They select and construct appropriate representations to organise data and information. They process data and information and analyse it to describe patterns, trends and relationships. They identify possible sources of error in methods and identify unanswered questions in conclusions and claims. They identify evidence to support their conclusions and construct arguments to support or dispute claims. They select and use language and text features appropriately for their purpose and audience when communicating their ideas and findings.
Source: Australian Curriculum, Assessment and Reporting Authority (ACARA), *Australian Curriculum Version 9.0 Science for Foundation–10* https://v9.australiancurriculum.edu.au/f-10-curriculum/learning-areas/science/year-7

Year 7 Science standard elaborations

		A	B	C	D	E
		The folio of student work contains evidence of the following:				
Science understanding	**Biological sciences**	reasoned explanation of how biological diversity is ordered and organised	informed explanation of how biological diversity is ordered and organised	explanation of how biological diversity is ordered and organised	description of the order or organisation of biological diversity	statement/s about ordering biological diversity
		• purposeful representation of flows of matter and energy in ecosystems • reasoned prediction of the effects of environmental changes	• informed representation of flows of matter and energy in ecosystems • plausible prediction of the effects of environmental changes	• representation of flows of matter and energy in ecosystems • prediction of the effects of environmental changes	• partial representation of flows of matter and energy in ecosystems • description of the effects of environmental changes	• directed representation of flows of matter and energy in ecosystems • statement/s about environmental changes

		A	B	C	D	E
	Earth and space sciences	• purposeful modelling of cycles in the Earth-sun-moon system • reasoned explanation of the effects of these cycles on Earth phenomena	• informed modelling of cycles in the Earth-sun-moon system • informed explanation of the effects of these cycles on Earth phenomena	• modelling of cycles in the Earth-sun-moon system • explanation of the effects of these cycles on Earth phenomena	• partial modelling of cycles in the Earth-sun-moon system • description of the effects of these cycles on Earth phenomena	• modelling of cycles in the Earth-sun-moon system, with direction • statement/s about cycles on Earth
	Physical sciences	• purposeful representation of the effects of forces acting on objects • reasoned explanation of the effects of forces acting on objects	• informed representation of the effects of forces acting on objects • informed explanation of the effects of forces acting on objects	• representation of the effects of forces acting on objects • explanation of the effects of forces acting on objects	• partial representation of the effects of forces acting on objects • description of forces acting on objects	• statement/s about forces
	Chemical sciences	• purposeful use of particle theory to explain the physical properties of substances • considered development of processes that separate mixtures	• effective use of particle theory to explain the physical properties of substances • effective development of processes that separate mixtures	• use of particle theory to explain the physical properties of substances • development of processes that separate mixtures	• description of particle theory • guided development of processes that separate mixtures	• statement/s about particle theory • directed development of processes that separate mixtures
Science as a human endeavour	**Nature and development of science**	considered identification of the factors that can influence development of and lead to changes in scientific knowledge	informed identification of the factors that can influence development of and lead to changes in scientific knowledge	identification of the factors that can influence development of and lead to changes in scientific knowledge	identification of the factors that can influence development of or lead to changes in scientific knowledge	statement/s about the development of scientific knowledge

		A	B	C	D	E
	Use and influence of science	reasoned explanation of how scientific responses are developed and can impact society	informed explanation of how scientific responses are developed and can impact society	explanation of how scientific responses are developed and can impact society	description of an impact of scientific responses on society	statement/s about scientific responses
		reasoned explanation of the role of science communication in shaping viewpoints, policies and regulations	informed explanation of the role of science communication in shaping viewpoints, policies and regulations	explanation of the role of science communication in shaping viewpoints, policies and regulations	description of the role of science communication in shaping viewpoints, policies and regulations	description of science communication
Science inquiry	Questioning and predicting	purposeful planning of investigations to test: • relationships • aspects of scientific models	plausible planning of investigations to test: • relationships • aspects of scientific models	planning of investigations to test: • relationships • aspects of scientific models	guided planning of investigations to test: • relationships • aspects of scientific models	directed planning of investigations to test: • relationships • aspects of scientific models
	Planning and conducting	thorough planning and conducting of safe, reproducible investigations	detailed planning and conducting of safe, reproducible investigations	planning and conducting of safe, reproducible investigations	planning and conducting of safe investigations	conducting of safe investigations
		considered identification of potential ethical issues and intercultural considerations required for field locations or use of secondary data	informed identification of potential ethical issues and intercultural considerations required for field locations or use of secondary data	identification of potential ethical issues and intercultural considerations required for field locations or use of secondary data	guided identification of potential ethical issues and intercultural considerations required for field locations or use of secondary data	directed identification of potential ethical issues and intercultural considerations required for field locations or use of secondary data
		use of equipment for the considered generation and recording of data with precision	use of equipment for the effective generation and recording of data with precision	use of equipment for the generation and recording of data with precision	use of equipment for the generation and recording of data	directed use of equipment for the generation and recording of data

		A	B	C	D	E
	Processing, modelling and analysing	selection and construction of appropriate representations for the purposeful organisation of data and information	selection and construction of appropriate representations for the effective organisation of data and information	selection and construction of appropriate representations for the organisation of data and information	selection and construction of representations for the organisation of data and information	use of provided representations for the organisation of data and information
		• thorough processing of data and information • thorough analysis of data and information to describe patterns, trends and relationships	• detailed processing of data and information • detailed analysis of data and information to describe patterns, trends and relationships	• processing of data and information • analysis of data and information to describe patterns, trends and relationships	• processing of data and information, with guidance • use of data and information to identify patterns, trends and relationships	statement/s about patterns, trends or relationships
	Evaluating	• considered identification of possible sources of error in methods • considered identification of unanswered questions in conclusions and claims	• informed identification of possible sources of error in methods • informed identification of unanswered questions in conclusions and claims	• identification of possible sources of error in methods • identification of unanswered questions in conclusions and claims	• guided identification of possible sources of error in methods • guided identification of unanswered questions in conclusions and claims	• statement/s about errors in methods • directed identification of unanswered questions in conclusions and claims
		• purposeful identification of evidence to support their conclusions • considered construction of arguments to support or dispute claims	• informed identification of evidence to support their conclusions • informed construction of arguments to support or dispute claims	• identification of evidence to support their conclusions • construction of arguments to support or dispute claims	• guided identification of evidence to support their conclusions • guided construction of arguments to support or dispute claims	• directed identification of evidence to support their conclusions • directed construction of arguments to support or dispute claims

		A	B	C	D	E
	Communicating	appropriate selection and purposeful use of language and text features, including scientific terminology, for their purpose and audience when communicating their ideas and findings.	appropriate selection and use of language and text features, including scientific terminology, for their purpose and audience when communicating their ideas and findings.	appropriate selection and use of language and text features for their purpose and audience when communicating their ideas and findings.	use of language and text features for their purpose and audience when communicating their ideas and findings.	use of language and text features when communicating their ideas and findings.

Key	shading emphasises the qualities that discriminate between the A–E descriptors

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